GIS Programming and Customization
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Income Distribution Toolbox Project Report

Introduction.

The overall goal of this workflow is to create a Python toolbox with a custom script tool in ArcGIS Pro that calculates the quantile of family incomes. The sub-goals of the script tool are to classify numerical data into different categories by manipulation of the Excel data , then joining the attribute table of a geodatabase file with the Excel datasets, and creating a new feature layer based on the joined table that visualize the family income distribution.

Study area.

The study area is California, United States. California state is a western U.S. state that stretches from the Mexican border along the Pacific for nearly 900 miles. Its terrain includes cliff-lined beaches, redwood forest, the Sierra Nevada Mountains, Central Valley farmland and the Mojave Desert.

Data and Data Processing.

The dataset includes an excel file and a geodatabase file. The excel file contains the annual income in each county within California. The geodatabase file is the California county boundaries. This layer provides an initial offering as "best available" at 1:24,000 scale. The spatial reference is NAD 1983.

The original Excel data has one table sheet and includes six variables: County, Population, Population Density, Per capita income, Median household income, and Median family income. The columns of the County and Median family income chosen as the new Excel file for further analysis.

Methods.

The first step is to create a new python toolbox. The toolbox is created by clicking the “Toolboxes” then “New Python Toolbox” in the Catalog pane of ArcGIS Pro. In the Toolbox class' __init__ method, properties of the toolbox are defined, including the alias, label, and description. The name of the toolbox is defined by the name of the .pyt file.

As the code below, the second step is to manipulate the excel data. Use the DataFrame.quantile to calculate the Q1, Q2, Q3 and IQR of the family income. Based on the quantiles results, classify numerical data into six categories.Then, create new excel file by DataFrame.to_excel.

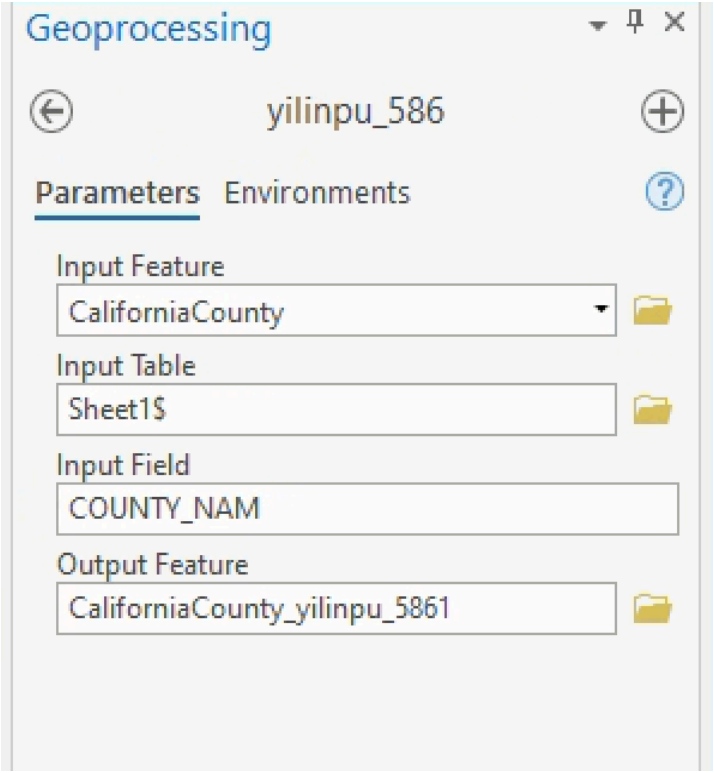
```
# Calculate IQR
Median = df[inputf3].quantile(0.5)
Q1 = df[inputf3].quantile(0.25)
Q3 = df[inputf3].quantile(0.75)
IQR = Q3 - Q1

# identify the statistics of data
for i in df[inputf3]:
    if i >= (Q3 + 1.5 * IQR):
        statistics.append('OutliersOverQ3')
    elif (Q1 - 1.5 * IQR) < i <= Q1:
        statistics.append('< 25%')
    elif Q3 <= i < (Q3 + 1.5 * IQR):
        statistics.append('> 75%')
    elif i <= (Q1 - 1.5 * IQR):
        statistics.append('OutliersBelowQ1')
    elif Median <= i < Q3:
        statistics.append('50%-75%')
    elif Q1 < i < Median:
        statistics.append('25%-50%')
    else:
        statistics.append('missing value')

df['statistics'] = statistics
str3 = str2[0:2]
df.to_excel(str3 + "/output.xlsx")
```

The third step is to join the attribute table of a geodatabase file and the excel file. The python code is arcpy.AddJoin_management(inFeatures, joinField, joinTable, joinField) that inFeatures is the geodatabase file, joinTable is the new Excel file. This line of code will call the Add Join tool in ArcGIS Pro to execute the join step. Based on the joined

table, create a new feature layer by arcpy.CopyFeature_management(joined_table, outputfc).

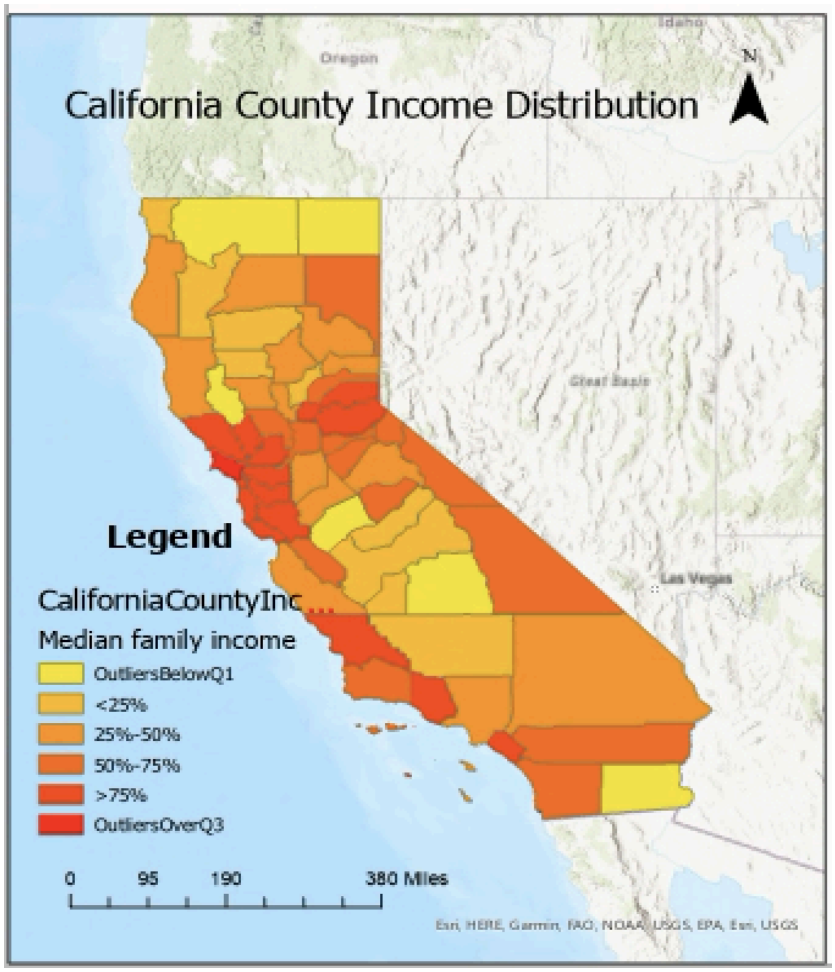


Results.

The results of toolbox generates the statistics quantile to classify the numerical data from Excel file into six different categories: outliers over Q3, less than 25%(Q1), greater than 75%(Q3), outliers below Q1, 25%-50%(Q1-Q2), and 50%-75% (Q2-Q3). Then use AddJoin and Copyfeature functions to join the Excel table and the attribute table of a geodatabase file and create a new feature layer based on the joined table.

The figure in the left is the toolbox created, the input feature is the geodatabase file CaliforniaCounty, the input table is the excel file California income by county, the input field COUNTY_NAM is the name of each county.

After running the toolbox that joins the geodatabase file of California County Boundaries and the excel file of California median family income by the field COUNTY_NAM. The new feature layer with classification visualized in the map below.



Discussion.

I use the statistical knowledge of quantiles to classify the categories of income data. Using the quantiles class can grouped the numerical data into minimum, Q1, Q2 (median), Q3, and maximum. This method directly shows the statistical range distribution of the sample dataset and visualizes the income distribution within California state by county. The limitation of the quantiles method is that it is not affected by extreme values. Therefore, the class of outliers categorized as Lower Outliers and Higher Outliers as the legend in the figure above.

The code can be reused to calculate and visualize the income distribution of any state. Or it can be reused to calculate the distribution of population of any state or country.

References.

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